

## SAFETY DATA SHEET

Revision Date 01-April-2024

Revision Number 3

### 1. Identification

**Product Name** Polystyrene latex microsphere, 1 micron, 2.5 wt% dispersion in water

**Cat No. :** 42743

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Not classified under the Hazardous Products Regulations (SOR/2015-17)

Based on available data, the classification criteria are not met

#### Label Elements

None required

#### **Hazard Statements**

#### **Precautionary Statements**

### 3. Composition/Information on Ingredients

| Component | CAS-No    | Weight % |
|-----------|-----------|----------|
| Water     | 7732-18-5 | 97.5     |

|             |           |     |
|-------------|-----------|-----|
| Polystyrene | 9003-53-6 | 2.5 |
|-------------|-----------|-----|

#### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                            |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                                                                                              |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.                                                                                                                                                                                                                                                                                      |
| <b>Inhalation</b>                      | Remove to fresh air. Get medical attention immediately if symptoms occur.                                                                                                                                                                                                                                                                                                                                    |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                                                                                                                                                                                                                                                                                        |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                        |

#### 5. Fire-fighting measures

|                                         |                                                                                                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Not combustible. CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                             |
| <b>Flash Point</b>                      | No information available                                                                                                             |
| <b>Method -</b>                         | No information available                                                                                                             |
| <b>Autoignition Temperature</b>         | No information available                                                                                                             |
| <b>Explosion Limits</b>                 |                                                                                                                                      |
| <b>Upper</b>                            | No data available                                                                                                                    |
| <b>Lower</b>                            | No data available                                                                                                                    |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                             |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                             |

#### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. None reasonably foreseeable. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 3                   | 0                  | -                       |

## 6. Accidental release measures

### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required.

### Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal.

## 7. Handling and storage

### Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

### Storage.

Keep refrigerated. Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Incompatible Materials. Oxidizing agent.

## 8. Exposure controls / personal protection

### Exposure Guidelines

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | 480 minutes       | 0.11mm          | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Multi-purpose/ABEK conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                                                        |
|----------------------------------------|--------------------------------------------------------|
| Physical State                         | Liquid dispersion                                      |
| Appearance                             | No information available                               |
| Odor                                   | Odorless                                               |
| Odor Threshold                         | No information available                               |
| pH                                     | No information available                               |
| Melting Point/Range                    | No data available                                      |
| Boiling Point/Range                    | No information available                               |
| Flash Point                            | No information available                               |
| Evaporation Rate                       | No information available                               |
| Flammability (solid,gas)               | Not applicable                                         |
| Flammability or explosive limits       |                                                        |
| Upper                                  | No data available                                      |
| Lower                                  | No data available                                      |
| Vapor Pressure                         | <=1100 hPa @ 50 °C                                     |
| Vapor Density                          | No information available                               |
| Specific Gravity                       | No information available                               |
| Solubility                             | No information available                               |
| Partition coefficient; n-octanol/water | No data available                                      |
| Autoignition Temperature               | No information available                               |
| Decomposition Temperature              | No information available                               |
| Viscosity                              | No information available                               |
| Molecular Formula                      | [-CH <sub>2</sub> CH(C <sub>6</sub> H <sub>5</sub> )-] |

## 10. Stability and reactivity

|                                  |                                                                   |
|----------------------------------|-------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                        |
| Stability                        | Stable under normal conditions.                                   |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Oxidizing agent                                                   |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )           |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                          |
| Hazardous Reactions              | None under normal processing.                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------|
| Oral LD50   | Category 3. ATE = 50 - 300 mg/kg. Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.   |
| Dermal LD50 | Category 3. ATE = 200 - 1000 mg/kg. Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Vapor LC50  | Category 3. ATE = 2 - 10 mg/l. Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.         |

#### Component Information

| Component | LD50 Oral | LD50 Dermal | LC50 Inhalation |
|-----------|-----------|-------------|-----------------|
|-----------|-----------|-------------|-----------------|

|       |   |   |   |
|-------|---|---|---|
| Water | - | - | - |
|-------|---|---|---|

**Toxicologically Synergistic** No information available

**Products**

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component   | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------|-----------|------------|------------|------------|------------|------------|
| Water       | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Polystyrene | 9003-53-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

**Persistence and Degradability** Persistence is unlikely Miscible with water based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT** Not regulated

**TDG** Not regulated

**IATA** Not regulated

**IMDG/IMO** Not regulated

## 15. Regulatory information

### International Inventories

| Component   | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP       |
|-------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----------|
| Water       | 7732-18-5 | X   | -    | X    | ACTIVE                                        | 231-791-2 | -      | -         |
| Polystyrene | 9003-53-6 | X   | -    | X    | ACTIVE                                        | -         | -      | 500-008-9 |

| Component   | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Water       | 7732-18-5 | X     | KE-35400 | X    | -    | X    | X    | X     | X     |
| Polystyrene | 9003-53-6 | X     | KE-13257 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Legend

NPRI - National Pollutant Release Inventory

### Other International Regulations

Authorisation/Restrictions according to EU REACH

Not applicable

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component   | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Water       | 7732-18-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Polystyrene | 9003-53-6 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component   | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water       | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Polystyrene | 9003-53-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

Prepared By

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[www.thermofisher.com](http://www.thermofisher.com)

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|-------------------------|----------------------------------------------------|
| <b>Revision Date</b>    | 01-April-2024                                      |
| <b>Print Date</b>       | 01-April-2024                                      |
| <b>Revision Summary</b> | New emergency telephone response service provider. |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**